MILITARY SPECIFICATION
FOR
FUZE, ELECTRONIC TIME:
M724

This specification is approved for use by USA Armaments Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the FUZE, ELECTRONIC TIME: M724 for use on cargo carrying artillery projectiles. This fuze will be used with Fuze Setter: M36.

1.2 Description. The M724 is a digital electronic artillery time fuze. It consists of an electronic timer, a safety and arming mechanism, and a chemical reserve power supply. The overall length of the fuze is approximately 5.27 inches; the weight is approximately 1.70 pounds. There are two modes of fuze operation: point detonation (PD); and electronic time (ET) with limited PD back-up. The M724 is used in the 105mm howitzer, 155mm howitzer, and 8 inch howitzer.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents and documents cited therein of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which maybe of use in improving this document should be addressed to: USA Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ, 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Military

MIL-A-2550 -- Ammunition and Special Weapons, General Specification for

MIL-S-46844 -- Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type

MIL-F-48701 -- Fuze Setter: M36

MIL-S-50827 -- Soft Soldered Electrical Connections for Conventional Weapons Including Other Related Electronic Devices

MIL-Q-50829 -- General Quality Assurance Provisions for Proximity Fuzes and Related Components

MIL-P-60412 -- Packaging, Packing and Marking for Shipment of Artillery type and Rocket Fuzes, General Specification For

STANDARDS

Military

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-331 -- Fuze and Fuze Components, Environmental and Performance Tests for

MIL-STD-1167 -- Ammunition Data Card

MIL-STD-1168 -- Lot Numbering of Ammunition

MIL-STD-1235 -- Single and Multiple Continuous Sampling Procedures and Tables for Inspection by Attributes

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DRAWINGS

USA, Harry Diamond Laboratories (FSCM 19202)

PL 11711268 -- FUZE, Electronic Time: M724

PDL 11711268 -- FUZE, Electronic Time: M724, Packaging For

EDL 11711268 -- Equipment Data List for FUZE, Electronic Time: M724

11720206 -- Cup, Booster

11720213 -- Pellet, Booster

11722634 -- Packaging and Packing for Rear Fitting

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49 CFR 171-190 -- Department of Transportation Rules and Regulations for the Transportation of Hazardous Materials

(Request for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Samples. The contractor is responsible for all requirements for providing and testing of samples as prescribed herein. When this specification is assigned to a subcontractor, it shall be the

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prime contractor's responsibility to insure that sample requirements are met. The prime contractor shall notify the Government at the time of assignment to a subcontractor whether the testing will be performed by the prime or by the subcontractor, as well as the method of control by the prime contractor.

3.1.1 First article approval sample. Prior to the start of regular production, except when production on a new contract at the same facility follows within 90 days production of acceptable material under this specification, the contractor shall manufacture and submit a first article approval sample of parts and subassemblies specified in Table 1 and Figure 1. The first article approval sample shall be produced using as far as possible the methods and processes proposed for quantity production. When processes differ from the production processes, they shall be basically equivalent and the difference in process and resultant product shall be fully described in documentation submitted with the first article sample. The sample shall conform to the requirements of this specification. Additional samples that may be required because of failure of the sample to meet the requirements of this specification shall be supplied by the contractor at his expense. Prior to approval of the first article approval sample, acquisition of parts and materials or initiation of production will be at the sole risk of the contractor. Any first article sample tests conducted by the contractor must be controlled and witnessed by a qualified representative of the Government procuring activity.

3.1.2 Supplemental samples. At the Government's discretion, a supplemental sample may be required whenever there is a break in production continuity of 90 days or more, or a change in design, material or process that may affect safety, operability, reliability, or interchangeability as defined by the Performance Requirements (3.3) and Classification of Defects (4.4.5). In all cases, such changes will require prior approval of the procuring activity.

3.1.3 Comparison sample. Thirty-two fuzes selected at random from each accepted lot, or as specified by the Government Procuring Activity, shall be shipped for evaluation within five working days after lot acceptance to the technical agency designated by the procuring activity. (See 6.1.3)

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3.2 Construction. The device shall be constructed in accordance with the applicable drawings (PL 11711268) to the extent prescribed herein.

3.2.1 Materials. Materials shall be those specified by the applicable drawings.

3.2.2 Dimensions and technical notes

3.2.2.1 Listed. Those dimensions and technical requirements listed in the Classification of Defects (4.4.5) are mandatory.

3.2.2.2 Unlisted. The contractor may propose deviations from characteristics shown on the drawings but not listed in the Classification of Defects for the purpose of adapting the item to established manufacturing practices. Such proposals must be accompanied by evidence that the deviation does not affect the function of the item and that all requirements will be met. If the Government confirms the contention of the contractor, the deviation will be approved for the duration of the contract. In case of dispute, the characteristics of the drawings shall apply. Approval of a deviation under provisions of this paragraph does not relieve the contractor from establishing and maintaining an adequate quality assurance program as elsewhere required. Provisions of this paragraph shall not be used to obtain approval for use of discrepant material (i.e., nonconforming material produced before approval is obtained); or for design changes, which should be requested in accordance with change provisions of the contract document. It is expected that deviations will normally be approved prior to the construction of the First Article Approval Sample and be reflected therein.

3.2.2.3 Interchangeability. While the dimensions and tolerances shown in the drawings and specifications will generally insure satisfactory products, the contractor is notified that some selection or matching may be required to obtain combinations which provide proper function or economic use of components. The contractor is responsible for selecting combinations of tolerances within the specified limits which satisfy his process needs and also meet the specified performance and fit requirements.

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3.3 Performance requirements. The performance requirements contained herein are mandatory. The definitions of terms contained in 6.4 shall apply.

3.3.1 Safety. The fuze shall not cause a bore burst or muzzle burst and shall not be capable of initiating function within 400 calibers (minimum mechanical arming distance) of the cannon.

3.3.2 Ballistic functioning. The fuzes, when set for electronic time (ET) by Fuze Setter M36 , shall function at the set time within the fuze set time tolerance if that maximum time occurs before reaching the end of the trajectory. The fuzes shall function when fired over the temperature range of -40°C to +63°C (-40°F to +145°F). The fuzes when set for time function, 4 seconds or greater, shall function on impact (PD) when impact occurs within 3 seconds prior to set time, if that impact occurs no less than 800 calibers from the muzzle. The fuzes, when set PD, shall function on impact, if that impact occurs no less than 800 calibers from the muzzle. The fuze shall meet the safety requirements of 3.3.1.

3.3.2.1 Premature bursts and early bursts. There shall be no premature bursts or early bursts in the gun tube or in flight.

3.3.2.2 Target impact non-function distance. The fuze, when set to PD by Fuze Setter M36 shall not cause the projectile to function upon impact of the fuze when fired in the 105mm, M103 Tube, Charge 7 at 8 ±1 inch thick plywood target located 42 - 1.0 meters from the muzzle of the weapon.

3.3.2.3 Target impact function distance. The fuze, when set to PD by Fuze Setter M36 , shall cause the projectile to function upon target impact when fired in the 155mm, M1 or M1A1 howitzer, charge 2, at an 8 ±1 inch thick plywood target located 124 + 1.5 meters from the muzzle of the weapon.

3.3.2.4 Electronic time functioning. The fuze when set to a time by Fuze Setter M36 shall cause the projectile to function airburst within ±0.3 seconds of set time with an overall root mean square error not exceeding 0.1 seconds.

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3.3.3 Electrical requirements.

3.3.3.1 Set and interrogate. The fuze and the electronics and nose cone assembly (11711430) shall be capable of being set either to Point Detonation (PD) or to a time from 0.2 to 199.9 seconds in 0.1 second increments by Fuze Setter M36. When interrogated by Fuze Setter M36, the interrogation results displayed on the setter shall be:

Setting	Interrogation Results
PD	P
0.3 seconds	0.29 to .33 inclusive
4.0 seconds	3.94 to 4.08 inclusive
25.0 seconds	24.94 to 25.08 inclusive
50.0 seconds	49.94 to 50.08 inclusive
68.9 seconds	68.84 to 68.98 inclusive
75.0 seconds	74.94 to 75.08 inclusive
100.0 seconds	99.94 to 100.08 inclusive
148.0 seconds	147.94 to 148.08 inclusive
199.9 seconds	199.84 to 199.98 inclusive

For interrogation of set times not shown in the table above, the upper and lower limits shall be interpolated.

3.3.3.1.2 If enough fuzes or electronics assemblies error on setting or interrogation (or display the wrong results on interrogation) to cause the lot to fail, then the failed units shall be checked three times each with two additional setters. If proper results are obtained with both additional setters, then the fuzes shall be considered to pass and the setter originally used shall be removed from use for acceptance testing and replaced. See 4.5.2.4.

3.3.3.1.3 During the set or interrogate cycle, there shall be a maximum of 0.1 volts (positive or negative) output at terminal J2-4.

3.3.3.1.1 Any fuze or electronics assembly for which the setter displays an "E" (for error) on setting or interrogation shall be tryed three additional times. If all three subsequent operations meet the requirements of this paragraph, the initial attempt shall be ignored. This provision shall not be used to pass a fuze or electronics assembly which operates intermittently. Multiple operations in the hopes of getting three proper results in a row is not permitted.

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3.3.3.2 Electronics operation. The electronics and nose cone assembly shall meet 3.3.3.1 and the following requirements when tested over the temperature range of -45°C to +68°C (-49°F to +155°F).

3.3.3.2.1 Setting with parallel loads. The electronics shall be capable of being set by Fuze Setter M36 (3.3.3.1) with a parallel load of 3.3 kilohm maximum resistance between any or all setting rings (J1).

3.3.3.2.2 Input voltage. The electronics shall meet all functional requirements on any supply voltage from -1.2 to -1.95 volts.

3.3.3.2.3 Current drain. The electronics shall draw a maximum of 920 milliamperes (ma) d.c. from a -1.95 volt minimum source (See 6.4.14 for a definition of "minimum" in this context.)

3.3.3.2.4 Ramp voltage input. The electronics, when set to a time and powered by an input voltage in the form of a linear ramp of -5.0 (±5% at any point in time) volts per second up to -1.95 volts minimum, shall start counting by the time the ramp voltage reaches -1.2 volts, and shall function properly with the first scaler output considered zero time.

3.3.3.2.5 Scaler period. The full cycle period of the scaler output shall be 98 to 110 milliseconds (ms).

3.3.3.2.6 Electrical arming. For the electronics set to a time, the time between the start of firing capacitor charging and the fire pulse shall be 196 to 220 ms for set times equal to or less than 3.2 seconds and 3.33 to 3.74 seconds for set times equal to or greater than 3.8 seconds. For set times of 3.3 to 3.7 seconds, the time between the start of capacitor charging and the fire pulse shall be within one of the above sets of limits.

3.3.3.2.7 Fire pulse. The detonator fire pulse shall be 1000 ergs minimum into a load consisting of a 2.0 +0.1 ohm resistor.

3.3.3.2.8 Timing accuracy. The fire pulse shall occur within the limits given below for any of the input voltage conditions specified above. For step applied voltages, the time shall be measured from the application of power. For the ramp voltage (3.3.3.2.4), the time shall be measured from the first scaler output.

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Set Time (Seconds)	Function Time (Seconds)
0.3	0.29 to 0.33
4.0	3.90 to 4.10
199.9	199.80 to 200.00

3.3.3.2.9 PD operation. The electronics when set PD and powered by -1.2 volts maximum, shall charge the firing capacitor to a minimum of -21.0 volts. This requirement is intended only to verify that the fire pulse would meet the requirements of 3.3.3.2.7 if the impact switch were closed. The contractor may, at his option, use as an alternate requirement 0.2 volts less than the peak (negative) capacitor voltage obtained in a time out test of the same unit powered at -1.2 volts maximum, provided that unit met the requirements of 3.3.3.2.7 in that test. For example, hypothetical test unit number 26, when tested against this paragraph, gave a capacitor voltage of only -20.5 volts. However, in a time-out test at -1.2 volts the peak capacitor voltage for unit number 26 was -20.7 volts and the fire pulse met the requirements of 3.3.3.2.7. Unit number 26 would therefore meet the requirements of this paragraph.

3.3.4 Explosive train functioning. The explosive train, with a 0.130 inch minimum spacer located between the S&A module lead charge and output lead charge and a 0.130 inch minimum spacer located between the S&A module and detonator block, and when initiated electrically, shall function high order. High order detonation shall be defined as sufficient output from the output lead charge to produce a dent 0.040 inches deep minimum in a 2024-T4 or 2024-T351 aluminum witness block located flush with the rear face of the fuze sleeve. During ballistic testing, the explosive train shall cause high order functioning of the booster pellet (11720213) when fired on an HE projectile or proper initiation of the expelling charge or self registration charge when fired on a cargo projectile.

3.3.5 Environmental requirements. The fuze (or specified part or subassembly) shall meet the requirements of the following tests in MIL-STD-331.

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3.3.5 Environmental requirements. The fuze (or specified part or subassembly) shall meet the requirements of the following tests in MIL-STD-331.

3.3.5.1 Jolt and jumble. Test 101 followed by 102. The S&A module shall meet the requirements of 3.3.8 following this test.

3.3.5.2 Forty-foot drop. Test 103, except in a standard wooden overpack condition using wire-bound box(es) containing the specified number of fuzes and simulated fuzes to fill the remainder of the box(es). The rotor in the S&A module shall be in the fully safe position and locked by both spin detents following this test.

3.3.5.3 Temperature and humidity. Test 105. The S&A module shall meet the requirements of 3.3.8 and the electronics and nose cone assembly shall meet the requirements of 3.3.3 following this test.

3.3.5.4 Salt fog. Test 107, 48 hour procedure. There shall be no more than five (5) pits or corrosion spots on the fuze with no pit or spot greater than 1/16 inch in any dimension. Following this test, the electronics shall be capable of being set by Fuze Setter XM36E1, with no cleaning of the setting rings allowed except for wiping with a clean, dry cloth.

3.3.5.5 Five-foot drop. Test 111, procedure 2, criteria A. The rotor in the S&A module shall be in the fully safe position and locked by both spin detents. Following this test, the electronics and nose cone assembly shall meet the requirements of 3.3.3 unless the setting rings are too damaged to fit Fuze Setter M36

3.3.5.6 Thermal shock. Test 113. The fuze shall meet the requirements of 3.3 following this test.

3.3.5.7 Leak detection. Test 118, procedure 5.4 (bubble method). In addition to the requirements stated in Test 118, the fuze shall meet the requirements of 3.3 following this test.

3.3.5.8 Transportation vibration. Test 119, procedure 2. The fuze shall meet the requirements of 3.3 following this test.

3.3.6 Crimp joint. The crimp joint of a fuze assembled without the key installed in the fuze sleeve shall meet the requirements of 11711268.

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3.3.7 Potting porosity. The maximum void size, when determined by sectioning of the electronics and nose cone assembly, shall be no larger than specified on 11711430.

3.3.8 S&A module safe position. The rotor in the S&A module shall be in the fully out of line condition and locked by both spin detents and the setback pin.

3.4 Workmanship. All parts shall be manufactured and finished in a thoroughly workmanlike manner to insure satisfactory functioning and durability. (See MIL-A-2550, Condition of Materials, Parts and Assemblies).

3.4.1 Soldering. Soldering shall be in accordance with MIL-S-50827 or MIL-S-46844.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor quality assurance system. The contractor shall provide and maintain a quality assurance system in compliance with MIL-Q-50829.

4.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification in compliance with MIL-Q-50829.

4.3 First article approval sample. The first article approval sample (FAAS) as specified in 3.1.1 shall be examined and tested in accordance with the first article approval sample flow chart (Figure 1) Classification of Defects (4.4.5), and Material and

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Component Certification (4.4.7). Unless otherwise specified herein or by the contracting officer, the tests shall be performed by the contractor under the observation of the Government QA representative and a representative of the cognizant technical agency. The contractor shall supply complete data in reproducible form on the contractor tests and examination of the first article approval sample units. The data and tested samples shall be delivered to the cognizant technical agency.

4.3.1 Approval of first article sample. If the sample passes the criteria of specified examinations and tests, it shall be approved. If the sample fails in any of the examinations and tests, the failure analysis will be conducted by the contractor under Government surveillance. The results of the tests and failure analysis of the units, together with the engineering analysis of the units, shall form a basis for corrective action. Depending upon the degree of corrective action deemed necessary by the Government, the first article approval sample may be:

a. Accepted, provided recommended changes are made by the contractor prior to start of regular production or during production of the first lot as prescribed by the Government. Such portions of first article inspection needed to verify corrective action may be included in the acceptance requirements of the lot, at the Government's discretion.

b. Rejected, and new samples required for the tests failed and for related tests which might be invalidated by the corrective action.

c. Rejected, and a complete new sample required for approval.

In all cases, the contractor shall comply with any required changes for the duration of the contract.

4.3.1.1 Reinstitution of tests. Acceptance of the first article approval sample does not relieve the contractor from meeting all requirements of this specification throughout the contract. The Government reserves the right to independently verify all requirements by repeating first article approval sample tests or any portion thereof on any production lot. If such verification

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indicates failure to meet requirements previously verified by first article approval sample tests, the contractor will be required to take corrective action. If the requirements are not met on the first lot to which the correction can be applied, the Government reserves the right to institute tightened inspection on a lot basis by adding to the normal lot acceptance inspections all or such parts of the first article approval sample tests needed to verify corrective action. Normal acceptance inspection will be resumed after two consecutive lots have satisfactorily passed the tightened conditions.

4.3.2 Ballistic acceptance criteria for the first article sample. The ballistic acceptance criteria for the first article sample is as follows:

a. Safe position - Critical defect - Any fuze found with the rotor in the S&A module not in the full safe position and locked by both spin detents shall be cause for rejection.

b. Premature burst - Critical Defect - Any premature burst (see 6.4.4) shall be cause for rejection.

c. Function - Critical Defect - Any function (see 6.4.8) in the non-function test shall be cause for rejection.

d. Early burst - Critical Defect - Any early burst (see 6.4.5) shall be cause for rejection.

e. Point detonating dud - Major Defect - More than two point detonating duds (see 6.4.6) in the functioning test shall be cause for rejection.

f. Time improper - Major Defect - More than six time impropers total, or more than two time impropers (see 6.4.7) in any one time-set subgroup (see 6.4.3), shall be cause for rejection.

g. RMS error - Major Defect - A root mean square error (see 6.4.12) in any time-set subgroup greater than the limit specified in 3.3.2.4 shall be cause for rejection.

h. Setting at the proving ground. - Major Defect - More than five (5) fuzes total, or more than two fuzes in any environmentally preconditioned subgroup, which cannot be set by Fuze Setter M36 to the required setting (time or PD), or which interrogate outside the limits given in 3.3.3.1 shall be cause for rejection.

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4.4 Acceptance inspection. Inspection shall be as specified in MIL-A-2550 and in this document. The contractor shall notify the Government QA representative prior to proceeding with inspection for Classification of Defects (4.4.5) and Operational Tests (4.4.6). Inspection may be made at a higher assembly level provided that equal assurance provisions are satisfied, but not at a lower level than specified.

4.4.1 Lot formation. A lot shall consist of devices produced at one manufacturing location under the following conditions: continuous production; stabilized methods and techniques; and the same revision of drawings and specification except for those introduced without obsolescence.

4.4.2 Sampling. Sampling shall be in accordance with MIL-STD-105 and this document. Unless otherwise specified, inspection level II shall apply. Where appropriate and when approved by the Government the contractor may use MIL-STD-1235. When any other than the specified lot size, for inspection purposes, are agreed upon in conformance with the Military Standards MIL-STD-105 or MIL-STD-1235, appropriate AQL's for the new inspection lots shall be selected to give the same consumer's risk as the established AQL's for the original lot size.

4.4.3 Lot acceptance. Lots shall be accepted on the basis of Classification of Defects (4.4.5), Operational Tests (4.4.6) and Material and Component Certification (4.4.7).

4.4.4 Data recording. A written record shall be made of the results of all examinations and tests performed as specified elsewhere in section 4 of this specification. The test data shall be recorded on 80 column standard data processing punch cards, or other computer compatible input medium approved or specified by the Government. Data from each operating test shall be recorded, but only the defect summary need be recorded in the inspection of the classification of defects. Data layout of the card or other recording medium shall be as approved or specified by the Government. The completed cards shall be forwarded to the designated Government agency within two working days after the Government acceptance of the item or lot. If inspections are conducted on a sublot basis, the acceptance data for each sublot shall be so identified. (Engineering information, section 6).

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4.4.5 Classification of defects. The assembly shall be inspected by using the Classification of Defects and applicable drawings.

4.4.5.1 Fuze, ET: M724 (11711268)

Categories	Defects	AQL	Method of inspection
Critical	None Defined		
Major			
101	Error on set or interrogate, or .745 dia to center contact less than minimum distance.	0.65	Fuze Setter M36 and Gage
102	2.00-12-UNS-1A thread damaged	0.65	Gage
103	Nose cone improperly crimped to sleeve	0.65	Visual DS-1
104	Seating Shoulder to forward end exceeds max.	0.65	Gage
105	1.60-20-UNS-2A-LH thread damaged	0.65	Gage
106	Wrench slot max span or min width improper	0.65	Gage
107	Seating shoulder to aft end exceeds max.	0.65	Gage
108	Setting ring missing or damaged	0.65	Visual
Minor			
201	Lot No. or nomenclature omitted, illegible or improper	1.5	Visual
202	Fuze not set in PD mode for shipment	1.5	Fuze Setter M36

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4.4.5.2 Rear fitting (11720291).

Categories	Defects	AQL	Method of inspection
Critical			
1	S&A Module not in safe position	100% screening	4.6.7
Major			
101	Improper resistance pins + to -	0.65	Ohmmeter
102	Improper resistance pins + to T	0.65	Low current ohmmeter (See 4.5.2.3)

4.4.5.3 Bias spring (11720296).

Categories	Defects	AQL	Method of inspection
Critical	None Defined		
Major			
101	Firing pin erroneously present	0.65	Visual

4.4.5.4 Shipping and storage containers.

Categories	Defects	AQL	Method of inspection
Critical	None Defined		
Major	None Defined		
Minor			
201	Dessicant bag missing	1.5	Visual
202	Package marking omitted illegible or improper	4.0	Visual
203	Fuze missing	1.5	Visual

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4.4.5.5 Defect standards

DS-1. The nose cone shall be tight in the sleeve with the sleeve crimped over evenly against the nose cone for a full 360°. There shall be no evidence of removal of the cadmium plating nor any bare steel visible on the sleeve when viewed with the naked eye. Removal of the chromate coating exposing the cadmium plating is permitted.

4.4.6 Lot operating tests. Samples from each lot shall be subjected to the tests given below and in Figure 2. All tests, except for the ballistic firing portion of Figure 2, shall be performed by the contractor. When temperature conditioning is specified, it shall apply for each test in that section.

4.4.6.1 Electronics operating tests. Major defect.

Requirement	Requirement Paragraph	Test Paragraph	AQL
A. Ambient Operation	3.3.3.2	4.6.2.1	0.65
B. High Temperature Operation	3.3.3.2	4.6.2.2	1.5(S-4)
C. Low Temperature Operation	3.3.3.2	4.6.2.3	1.5(S-4)
D. Long Time-out	3.3.3.2	4.6.2.4	1.5(S-4)

4.4.6.2 Explosive train operating tests. Major defect.

Requirement	Requirement Paragraph	Test Paragraph	AQL
Explosive Train Output	3.3.4	4.6.3	0.40(S-4)

4.4.6.3 Environmental tests. Major defect.

Requirement	Requirement Paragraph	Test Paragraph	AQL
Salt Fog	3.3.5.4	4.6.4.4	1.5(S-4)

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4.4.6.4 Ballistic functioning. The lot acceptance sample will be tested by the Government in accordance with Figure 2 and the requirements of 3.3.1, 3.3.2, and 3.3.3.1. The acceptance criteria for the lot acceptance sample is as follows:

a. Safe position - Critical defect - Any fuze found with the rotor in the S&A module not in the full safe position and locked by both spin detents shall be cause for rejection of the lot.

b. Premature burst - Critical defect - Any premature burst shall be cause for rejection of the lot.

c. Function - Critical defect - Any function in the non-function test shall be cause for rejection of the lot.

d. Early burst - Critical defect - Any early burst shall be cause for rejection of the lot.

e. Point detonating dud - Major defect - More than two point detonating duds in the PD functioning test shall be cause for rejection of the lot.

f. Time improper - Major defect - More than five time impropers total, or more than two time impropers in any one time-set subgroup, shall be cause for rejection of the lot.

g. RMS error - Major defect - A root mean square error in any time-set subgroup greater than the limit specified in 3.3.2.4 shall be cause for rejection of the lot.

h. Setting at the proving ground - Major defect - More than five fuzes total, or more than two fuzes in any environmentally preconditioned subgroup, which cannot be set by Fuze Setter M36 to the required setting (time or PD) or which interrogate outside the limits given in 3.3.3.1 shall be cause for rejection of the lot.

4.4.6.5 Disposition of lot acceptance samples. Units in the sample which pass the non-destructive inspections and tests shall be returned to the lot. Any unit which contains one or more defects shall be rejected and set aside pending disposition instructions from the procuring activity.

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4.4.7 Material and component certification. Prior to acceptance under this procurement, the contractor shall demonstrate by means of certifications or statement of findings, that only materials and components conforming to the item specification and drawings, have been used.

4.5 Test conditions and equipment.

4.5.1 Test conditions. Unless otherwise specified, all measurements are to be made at normal ambient room conditions of temperature, relative humidity, and altitude. Temperature conversion between Fahrenheit and Celsius shall be carried to the nearest whole degree. The following table of tolerances shall be applied where applicable to specified test conditions unless otherwise specified in the test procedures:

a.	Temperature in test chamber	±5° F
b.	Relative humidity	±5 percent
c.	Vibration amplitude	±10 percent
d.	Shock acceleration	±10 percent

4.5.2 Inspection equipment. Inspection equipment shall be in accordance with the design level shown by EA 11711268 and of sufficient accuracy and quality to permit performance of the required inspection. Gage and instrument setting shall be a function of the equipment, and independent of the operator. Equipment shall be marked to indicate the limit of acceptance for each test. Test procedures and calibration procedures shall conform to MIL-Q-50829.

4.5.2.1 Electronics test equipment. The test equipment for operating tests of the electronics shall be interconnected as shown in Figure 3. The definitions of terms contained in 6.4 shall apply. All external connections to terminals J2-2 and J2-4 shall have a combined input impedance greater than 1 megohm resistance and less than 100 picofarads capacitance. All external connections to terminal J2-3 (except for the simulated detonator) shall have a minimum input resistance of 2000 ohms. Unless otherwise specified, all voltages are with respect to terminal J2-5 (ground). Instrument accuracy and data recording shall be as specified below.

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4.5.2.1.1 Instrument A. Fuze Setter M36 . Record entire display on each interrogate operation. See 4.5.2.4.

4.5.2.1.2 Instrument B. Peak-reading voltmeter. Record voltage to the nearest 0.01 volt.

4.5.2.1.3 Instrument C. Period meter. Record scaler period (10 period average) to the nearest 0.1 millisecond.

4.5.2.1.4 Instrument D. Time interval meter. Record time to the nearest 0.01 second.

4.5.2.1.5 Instrument E. Time interval meter. Record time to the nearest 0.01 second.

4.5.2.1.6 Instrument F. Time interval meter. Record time to the nearest 0.01 second.

4.5.2.1.7 Instrument G. Voltmeter (command hold). Record voltage at terminal J2-1 at time of first scaler output to nearest 0.01 volt.

4.5.2.1.8 Instrument H. Controlled or programable power supply, with integral current meter and voltmeter. Voltage shall be controlled at terminal J2-1 to the extent specified.

4.5.2.1.9 Instrument J. Energy meter. GO/NO GO data acceptable, provided that the acceptance limit is increased by the inaccuracy of the measurement.

4.5.2.2 MNOS and MOS handling and testing. Personnel and equipment grounding procedures should be used whenever handling or testing MOS or MNOS circuits, circuit card assemblies with such circuits, the unpotted electronics assembly, the electronics and nose cone assembly, or any other partial assembly having access to the terminals of these IC's.

4.5.2.3 Resistance measuring equipment. Inspection equipment used to measure the electric detonator resistance (pins "+" and "T" on rear fitting) shall be limited to a maximum test current (including surges) of 2 milliamperes (mA).

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4.5.2.4 Fuze setters. Fuze setters used for acceptance test equipment shall have been certified within the preceding twelve months to have passed the tests contained in Table 1 of MIL-F-48701.

4.6 Test procedures.

4.6.1 Ballistic functioning tests. The sample for ballistic testing shall be radiographically examined prior to shipment. The rotor in the S&A module shall be in the full safe position and locked by both spin locks. After arrival at the proving ground the fuze shall be assembled with a booster pellet (11720213) and cup (11720206) for all tests except for the test with M483A1 projectiles. Where temperature conditioning is indicated, the fuze, projectile and propellant charges shall be conditioned at that temperature for a minimum of four hours and fired within five minutes following removal from the temperature chamber. Inert-loaded projectiles may be used provided that, for other than the M483A1 projectile, a T-2 supplemental charge or equivalent is used to provide a positive indication of fuze functioning. Tests with the M483A1 projectile shall be in the fire-for-effect mode for cargo expulsion. The cargo may be inert loaded.

4.6.1.1 Time functioning. The fuze shall be set to the specified time using Fuze Setter M36. The specified times may be adjusted up or down twenty percent if necessary to meet firing range constraints. The fuze shall then be interrogated and both the set and interrogate times recorded. The fuze shall be fired as specified in Figure 1 or 2, as appropriate, and the time interval between muzzle exit and functioning measured to the nearest millisecond by three independant chronograph systems. All measurements shall be recorded and analyzed according to 4.6.8. If a round fails to function in flight, observation shall be made to determine if the round functions on impact. All defects shall be recorded. A backup system of clocks or stopwatches independent of any element of the chronograph system shall also be employed and their measurements recorded. The requirements of 3.3.1, 3.3.2 and 3.3.3.1 shall be met.

4.6.1.2 PD non-functioning. The fuze shall be set to PD and fired as specified in 3.3.2.2. The requirements of 3.3.1, 3.3.2 and 3.3.3.1 shall be met.

4.6.1.3 PD functioning. The fuze shall be set to PD and fired as specified in 3.3.2.3. The requirements of 3.3.1, 3.3.2 and 3.3.3.1 shall be met.

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4.6.2 Electronic functioning tests. The following tests shall be conducted on electronics and nose cone assemblies using the equipment and conditions specified in 4.5. When temperature conditioning is required, the unit shall be conditioned for two hours minimum prior to the test and during the test. If separate chambers are used for conditioning and testing, the transfer time shall be less than one minute. Data shall be recorded for each unit as specified in 4.5.2.1. For all tests, the load resistors are not applied across J1 and power is not applied to J2-1 unless specifically stated. When required, P1 is connected to J1 only until the display is lighted on instrument A and the display recorded. At no time should power be applied to J2-1 while P1 is connected to J1. In Tables 2 and 3, for any setting on instrument A other than the time-set mode (T), the digits on the time dial play no role and are expressed as "XXX.X" to indicate that any digits are acceptable. The entire series of sub-tests in either Table 2 or 3 shall be considered as one test for purposes of score. A unit is considered a failure if it fails one or more subtests, but is regarded as only one failure regardless of how many sub-tests it fails.

4.6.2.1 Ambient operation test. The units shall be subjected to the test specified in Table 2 at normal room ambient conditions. The requirements of 3.3.3.2 shall be met.

4.6.2.2 High temperature operation tests. The units shall be subjected to the test specified in Table 2 at the high temperature specified in 3.3.3.2. The requirements of 3.3.3.2 shall be met.

4.6.2.3 Low temperature operation test. The units shall be subjected to the test specified in Table 2 at the low temperature specified in 3.3.3.2. The requirements of 3.3.3.2 shall be met.

4.6.2.4 Long time-out test. The units shall be subjected to the test specified in Table 3 at normal room ambient conditions. The requirements of 3.3.3.2 shall be met.

4.6.3 Explosive train operating test. This test shall be conducted on rear fittings which have been removed from fuzes, or on a sub-lot basis on rear fittings drawn from each sub-lot of units. Unless otherwise specified this test shall be conducted at normal room ambient conditions. If temperature conditioning is specified, the entire test assembly shall be conditioned for two hours minimum and the test conducted within 60 seconds following removal from the conditioning chamber.

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4.6.3.1 Preparation. Short power supply pins "+" and "T" together. Remove the S&A module, bias spring, detonator block assembly, and power supply from the rear fitting, keeping the power supply plugged into the detonator block assembly. Discard the bias spring. The sleeve may be modified as necessary to perform the test. Arm the S&A module (M55 detonator positioned on the centerline of the module) and reinstall into the sleeve with the required thickness spacers above and below the S&A module. The spacers may be made of any material but shall have a 1.0 inch minimum inside diameter. Reinsert the power supply and detonator block assembly into the sleeve.

4.6.3.2 Firing. Remove the short across the power supply pins and initiate the electric detonator through pins "+" and "T". Any convenient electrical power source may be used to initiate the electric detonator.

4.6.3.3 Dent block and results. The dent block shall be a minimum of 1.5 inches in every dimension and shall be located against the output lead charge end of the sleeve. Each face may be used, but only for a single test. As an alternate, blocks 1 1/4 x 1 1/4 x 3/8 minimum may be used with each block used for a single test on one 1 1/4 x 1 1/4 face. Either type of block may be used in the form of long "sticks". For example, a block 5 x 1 1/4 x 3/8 could be used for four tests, spaced at 1 1/4" intervals (end tests 5/8 from edge) along one 1 1/4 wide face. Dent depth shall be read to an accuracy of 0.003 inch using a probe with a spherical tip radius of .02 to .03 inch. The requirements of 3.3.4 shall be met.

4.6.4 Environmental tests. Test numbers are from MIL-STD-331. All fuzes shall be radiographically examined for safe position of the S&A module rotor and spin detents before and after each test.

4.6.4.1 Jolt and jumble. Conducted on fully assembled fuzes in accordance with test 101 followed by Test 102, with radiographic examination following each test. The fuzes shall be dissassembled and evaluated. The requirements of 3.3.5.1 shall be met.

4.6.4.2 Forty-foot drop. Conducted on fully assembled fuzes in accordance with Test 103 as modified by 3.3.5.2. The fuzes shall be disassembled and evaluated. The requirements of 3.3.5.2 shall be met.

4.6.4.3 Temperature and humidity. Conducted on fully assembled fuzes in accordance with Test 105. The fuzes shall be disassembled and evaluated, including conducting the test specified in Table 2 on the electronics and nose cone assemblies. The requirements of 3.3.5.3 shall be met.

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4.6.4.4 Salt fog. This test shall be conducted according to test 107 (48 hour procedure) on electronics and nose cone assemblies which have been crimped to empty fuze sleeves, using the same tooling and setup as used for final crimping of fuzes. The output lead charge hole shall be sealed by any convenient method. The requirements of 3.3.5.4 shall be met.

4.6.4.5 Five-foot drop. Conducted on fully assembled fuzes in accordance with Test 111, procedure 2. The fuzes shall be disassembled and evaluated, including conducting the test specified in Table 2 on the electronics and nose cone assemblies. The requirements of 3.3.5.5 shall be met.

4.6.4.6 Thermal shock. Conducted on fully assembled fuzes in accordance with Test 113. The fuzes shall be shipped for ballistic testing following radiographic examination. The requirements of 3.3.5.6 shall be met.

4.6.4.7 Leak detection. Conducted on fully assembled fuzes in accordance with Test 118, procedure 5.4 (bubble method). The fuzes shall be shipped for ballistic testing following radiographic examination. The requirements of 3.3.5.6 shall be met.

4.6.4.8 Transportation vibration. Conducted on fully assembled fuzes in accordance with Test 119, procedure 2. The fuzes shall be shipped for ballistic testing following radiographic examination. The requirements of 3.3.5.6 shall be met.

4.6.5 Crimp joint. This test may, at the contractor's option, be performed on the units from the salt fog test. The units will consist of electronics and nose cone assemblies crimped to empty fuze sleeves using the same tooling and setup as used for final crimping of fuzes. The units shall not contain the key in the sleeve. Two flats shall be ground on the electronics and nose cone assemblies of the units with a span of 1 1/4 - 1/4 inch back to a distance 2 1/2 + 1/4 inch from the tip. The sleeve shall be gripped in a suitable fixture and a peak reading torque wrench (or equivalent) used to determine the torque at which the nose cone moves .020 inch with respect to the sleeve, measured at the gap between the nose cone and sleeve. The requirements of 3.3.6 shall be met.

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4.6.6 Potting porosity. This test may, at the contractor's option, be conducted on electronics and nose cone assemblies from the crimp joint test. The contractor will not be permitted to exercise this option to use a different set of units if a rejectable void is uncovered in grinding the flats for the torque test. In that event, the units from the crimp joint test shall be used for this test. The electronics and nose cone assemblies shall be sawed into a minimum of twelve parts such that no piece is greater than one inch in any dimension. At least two saw cuts shall be made perpendicular to a line through terminals J2-1 and J2-5 and parallel to the longest dimension of the nose cone. The exposed surfaces of the potting material shall be cleaned of all dust and loose particles and visually examined for presence and size of voids. The requirements of 3.3.7 shall be met.

4.6.7 S&A module safe position. The S&A module shall be tested by one or more probe stations after insertion into the fuze sleeve, but prior to insertion of the bias spring, for the presence and position of the rotor, both spin detents, and the setback pin. As an alternate the equipment used to pick the S&A module up for insertion into the fuze sleeve may be designed to not accept an S&A module in other than the specified position. This alternate may be used only if the pickup station maintains positive control of the S&A module such that only S&A modules picked up by that station can be inserted in sleeves. Design of the probe station(s), the contractor's plan for certification and periodic recertification of the probe station(s), and the measures to be taken to insure that only S&A modules accepted by this station are used in fuzes shall be approved by the cognizant technical agency prior to use. The requirements of 3.3.8 shall be met.

4.6.8 Analysis of time-to-airburst data. The raw data for time-to-airburst shall be analyzed by the following procedures.

4.6.8.1 Significant difference in chronographs. For units in which an observed airburst occurred, the chronograph times for each round shall be checked for significant differences (6.4.10) and lost times (6.4.11) and the results recorded. Lost time units are considered "no-test" for acceptance purposes.

4.6.8.2 Average function time. For each round not considered lost by the criteria of 6.4.10 or 4.6.11, the arithmatic average of the times to function from the different chronographs shall be determined.

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4.6.8.2 Off-times. The data shall be examined for off-time functions (6.4.9) and those units noted. These units are not used in the calculation of RMS error.

4.6.8.3 RMS error. The RMS error for the units not considered lost or off-time functions shall then be calculated as specified in 6.4.12. The RMS error shall be calculated for each sub-test (see 6.4.3).

4.6.8.4 Reporting. A summary shall then be compiled for each lot showing, for each sub-test, the number of rounds fired, the type of function (or dud) of each round, the number of lost times, and the RMS error for each sub-test. This summary shall then be compared to the requirements of 4.3.2 or 4.4.6.4, as appropriate.

5. PACKAGING

5.1 Rear fittings. If rear fittings are produced on subcontract or at another location of the prime contractor's, the use of packaging and packing in accordance with 11722634 is recommended. In any event, the packaging shall comply with 49 CRF, parts 171-190 for detonating fuzes (class C).

5.2 Fuzes. Fuze packaging, packing and marking shall be in accordance with MIL-P-60412.

5.3 Ammunition Data Cards. Ammunition data cards shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Items to be included in procurement documents.

6.1.1 Documents. Procurement documents should specify title, date and number of this specification.

6.1.2 Samples. The number of complete fuzes should be increased to include the samples as required in 3.1 and 4.4 since they are not included in the number of units required for delivery.

6.1.3 Government testing of comparison samples. These samples are subject to retesting by the Government to insure that first article approval requirements continue to be met. Failure of these samples will not affect the acceptance of the lot, but may indicate future mandatory action.

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6.1.4 Lead Time. Government testing of the first article approval sample as specified in Figure 1 will require a maximum of 60 days following delivery of the sample to the Government Proving Ground. If there are any failures, additional time will be required.

6.1.5 Data Requirements. The following data are required by this specification to be furnished to the Government and should be entered on DD Form 1423 if made a part of the contract:

First Article Test Results	4.3
Examination and Test Results in machine readible format	4.4.4
Ammunition Data Cards	5.3
MIL-Q-50829, para. 3.6.1 -- Contractor-furnished Designs	

6.2 Engineering information. Upon request, copies of the raw data collected in the acceptance of a lot will be made available to the Government representative. Additional data or data requiring special requirements, when needed for engineering purposes, will be specifically defined by the contract.

6.3 MIL-Q-50829. Copies of MIL-Q-50829 may be obtained from the Harry Diamond Laboratories, ATTN: Contracting Officer, 2800 Powder Mill Road, Adelphi, MD. 20783.

6.4 Definitions. See Figure 4.

6.4.1 Set. A fuze is set by fuze setter M36 by entering the desired setting on the switches of the setter and contacting the setter contacts to the rings on the nose of the fuze. See Draft Equipment Publication (DEP) 9-1390-525-14 for a complete description of setter operation.

6.4.2 Interrogate. A mode of operation whereby the fuze setter displays the setting of a previously set fuze without affecting that setting.

6.4.3 Ballistic sub-test. A group consisting of all fuzes fired from the same weapon, on the same projectile, at the same zone, at the same temperature, with the same preconditioning, and with the same specified set time. Changing the set time as provided in 4.6.1.1, even in the middle of a sub-test, shall not be considered to split the group into two sub-tests.

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6.4.4 Premature or permature burst. A malfunction in which a fuze functions before the required safe separation distance from the weapon. This includes functions in handling, loading and inside the barrel.

6.4.5 Early or early burst. A malfunction in which a time-set fuze functions prior to T-4.0 seconds (T is set time) but later than safe separation. For a PD-set fuze, a function prior to impact but later than safe separation.

6.4.6 Dud. A malfunction in which a ballistically fired fuze does not function at all, except where impact occurs prior to 800 caliber travel.

6.4.7 Time improper. A malfunction in which a time-set fuze does not function within 0.3 seconds of set time. This includes early off times, late off times, impacts and duds.

6.4.8 Ballistic function. A function is defined as initiation of the output lead charge, normally resulting in initiation of the projectile. For determining reliability, a function is proper only if the proper response (detonation or expulsion) of the projectile results. For determining safety, any function of the output lead charge at an unsafe time shall be considered a safety failure, whether or not normal functioning of the projectile results.

6.4.9 Off time. A malfunction in which a time-set fuze functions later than T-4.0 seconds, but not within 0.3 seconds of set time. Functions between T-4.0 and T-0.3 seconds are early off times, and functions in the air but later than T+0.3 seconds are late off times.

6.4.10 Significant difference in chronograph. When the highest or lowest chronograph reading differs by more than 0.01 seconds from the next closest chronograph reading, then that first reading shall be disregarded. If the remaining two chronograph readings do not agree within 0.01 seconds, then all three will be disregarded and the data considered lost.

6.4.11 Lost time. Whenever, for any reasons including 6.4.10, there are not two good chronograph readings for the function time, the time data shall be considered lost. This definition shall not be used to invalidate the results of a test round in which the observers report a malfunction (premature, early, off time, impact or dud).

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6.4.12 Root mean square error. The root mean square error of the chronograph results for each round in a sub-test shall be calculated using the following formula:

$$\text{RMS ERROR} = \frac{1}{N}\sqrt{\sum_{i=1}^{N}(T - X_i)^2}$$

Where N is the number of data points, X_i is the function time of each round and T is the set time of each fuze. Note that this formula may be used to calculate the RMS error for a sub-test even though the set time might not be identical for all rounds. (See 4.6.1.1 and 6.4.3).

6.4.13 Impact or impact function. A malfunction in which a time-set fuze functions on ground impact instead of in the air. If such an impact function occurs prior to T+0.3 seconds, the round shall be considered "no-test". Function on ground impact of a PD-set fuze is scored a proper.

6.4.14 Minimum and maximum. Whenever the words minimum or maximum are used in this specification to apply to a negative quantity, they shall be interpreted to apply to the absolute value of that quantity. For example, if -5.0 maximum is specified, -4.9 is acceptable and -5.1 is not. Similarily, -9.0 does not meet a requirement of -10.0 minimum.

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Table 1. FAAS Parts and Subassemblies

Submit a sample of 25 of each of the following parts and subassemblies. If any of these parts or subassemblies are replaced by another part number, the new part number shall be supplied. For bulk items, a quantity sufficient to produce 25 fuzes shall be supplied.

Part Number	Nomenclature
11711234	Diode, Zener
11711240	Capacitor, Ceramic
11711242	Diode, Zener
11711404-1	Capacitor, Tantalum
11711404-2	Capacitor, Tantalum
11711405-1	Resistor, Composition
11711405-2	Resistor, Composition
11711405-3	Resistor, Composition
11711406	Diode
11711407	Plug, Nose
11711408	Cone, Nose
11711409	Cover, Electronics
11711410	Cup, Orientation
11711411	Circuit Card No. 1
11711412	Circuit Card No. 2
11711413	Circuit Card No. 1 Assembly
11711414	Circuit Card No. 2 Assembly
11711417-1	Pad, Contact
11711417-2	Pad, Contact
11711418	Contact, Coil
11711419	Strip, Lead Frame
11711420	Bobbin Assembly
11711421	Wire, Core
11711422	Bobbin, Core
11711423	Core, Coil
11711424	Transistor, NPN
11711425	Setting Rings and Plug Assembly
11711428	Electronics Assembly
11711429	Resin, Epoxy
11711430	Electronics and Nose Cone Assembly
11711431	Hardner
11711444	Washer, Connector
11711445	Potting Compound
11711446	Mounting Board

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Table 1. FAAS Parts and Subassemblies (Con't)

Part Number	Nomenclature
11711447	Post, Mounting Board
11711448	Transformer, Encapsulated
11711450-1	Ring, Setting, Assembly
11711450-2	Ring, Setting, Assembly
11711478	Plug, Detonator Block
11718234	Clip, Detonator
11720214	Clip, Ground Pin
11720258	Output Lead
11720291-2	Rear Fitting
11720296-2	Spring, Bias
11720297	Contact, Detonator
11720298	Block, Detonator
11720299	Insulator, Contact, Detonator
11722620	Detonator Block Assembly
11722621	Clip, Shorting (If used)
11722622-2	Sleeve
11722630	Cap, Protective (If used)
11722636	Firing Lead and Power Supply Ass'y
11726804	Firing Lead Assembly
MS27183-4	Washer, Flat
MS59386-015	O-Ring
MS59386-033	O-Ring
NAS549-3	Washer, Fiber

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Table 2. Electronic Tests, Not Including Long Time-out

Test No.	Condition	Action	Instrument	Requirement
A.1	Load resistor J1-2 to J1-3.	Set A to T199.9; connect P1 to J1.	A	3.3.3.2.1
			B	3.3.3.1
A.2	Load resistor J1-1 to J1-2.	Set A to T068.9; connect P1 to J1.	A	3.3.3.2.1
A.3	Both resistors of A.1 and A.2	Set A to T148.0; connect P1 to J1.	A	3.3.3.2.1
A.4		Set A to T000.3; connect P1 to J1.	A	3.3.3.1
A.5		Set A to ?XXX.X; connect P1 to J1.	A	3.3.3.1
A.6	H set to provide -1.2 maximum volts at J2-1 after connection to J2-1.	Connect H to J2-1.	C	3.3.3.2.5
			E	3.3.3.2.8
			F	3.3.3.2.6
			J	3.3.3.2.7
A.7		Set A to T004.0; connect P1 to J1.	A	3.3.3.1
A.8		Set A to ?XXX.X; connect P1 to J1.	A	3.3.3.1
A.9	H set to provide ramp voltage (3.3.3.2.4) at J2-1 after connection to J2-1.	Connect H to J2-1.	C	3.3.3.2.5
			D	3.3.3.2.8
			F	3.3.3.2.6
			G	3.3.3.2.4
			H	3.3.3.2.3
			J	3.3.3.2.7
A.10		Set A to PXXX.X; connect P1 to J1.	A	3.3.3.1
A.11		Set A to ?XXX.X; connect P1 to J1.	A	3.3.3.1
A.12	H set to provide -1.2 maximum volts at J2-1 after connection to J2-1.	Connect H to J2-1.	B	3.3.3.2.9

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Table 3. Long Time-out Test

Test No.	Condition	Action	Instrument	Requirement
B.1		Set A to T199.9; connect P1 to J1.	A	3.3.3.1
B.2		Set A to ?XXX.X; connect P1 to J1.	A	3.3.3.1
B.3	H set to provide -1.95 minimum volts at J2-1 after connection to J2-1.	Connect H to J2-1	E F	3.3.3.2.8 3.3.3.2.6

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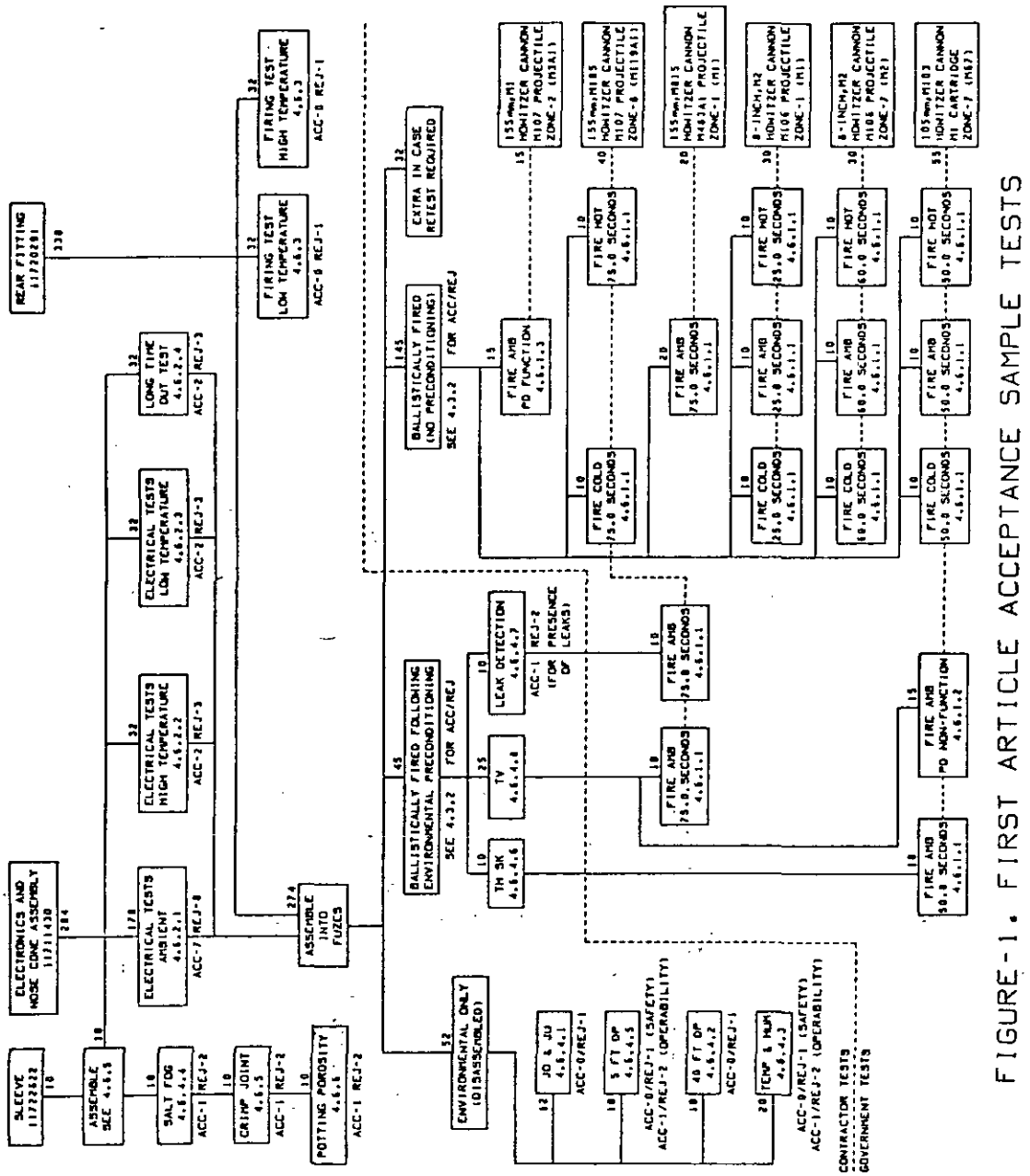


FIGURE-1. FIRST ARTICLE ACCEPTANCE SAMPLE TESTS

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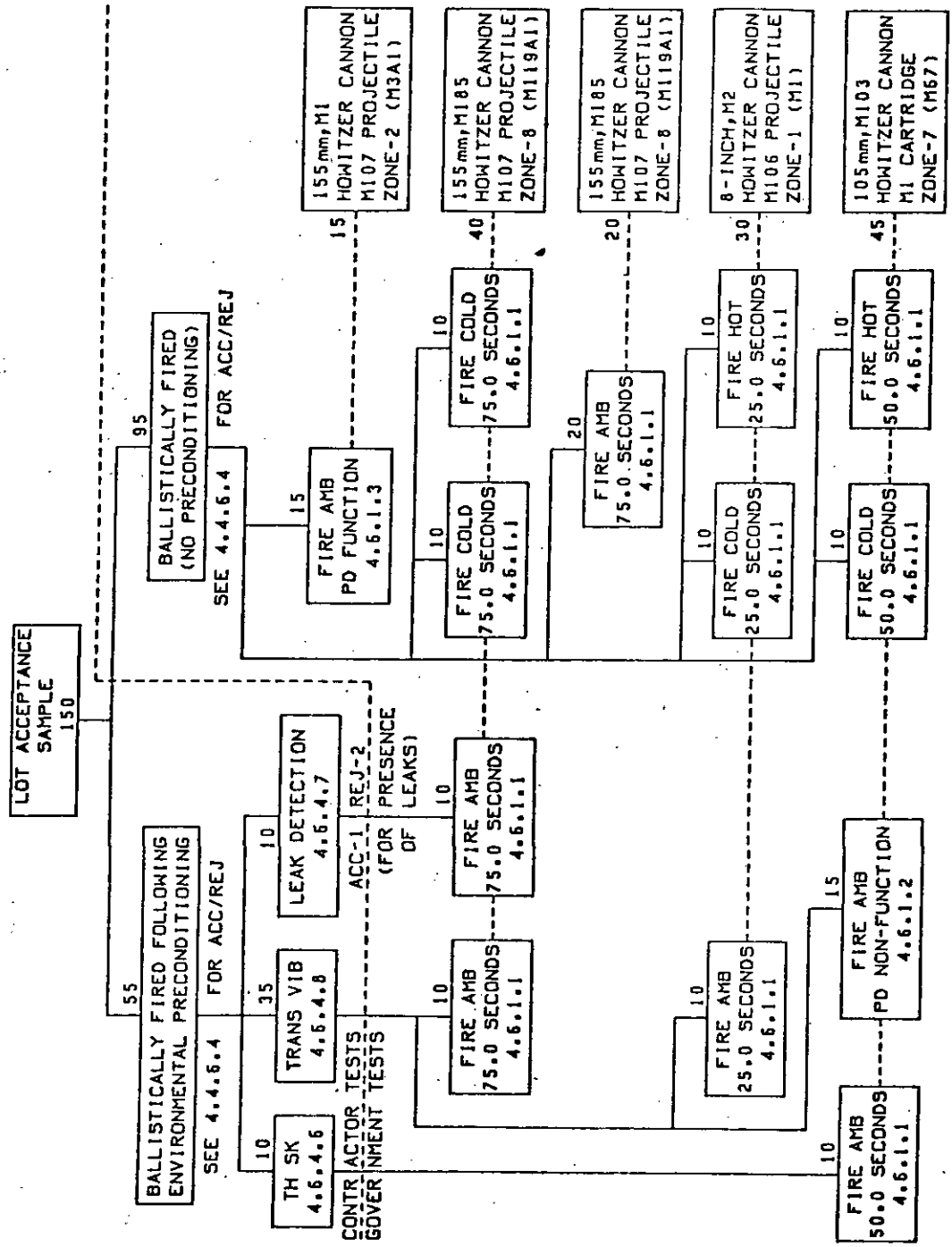


FIGURE-2. BALLISTIC LOT ACCEPTANCE TESTS

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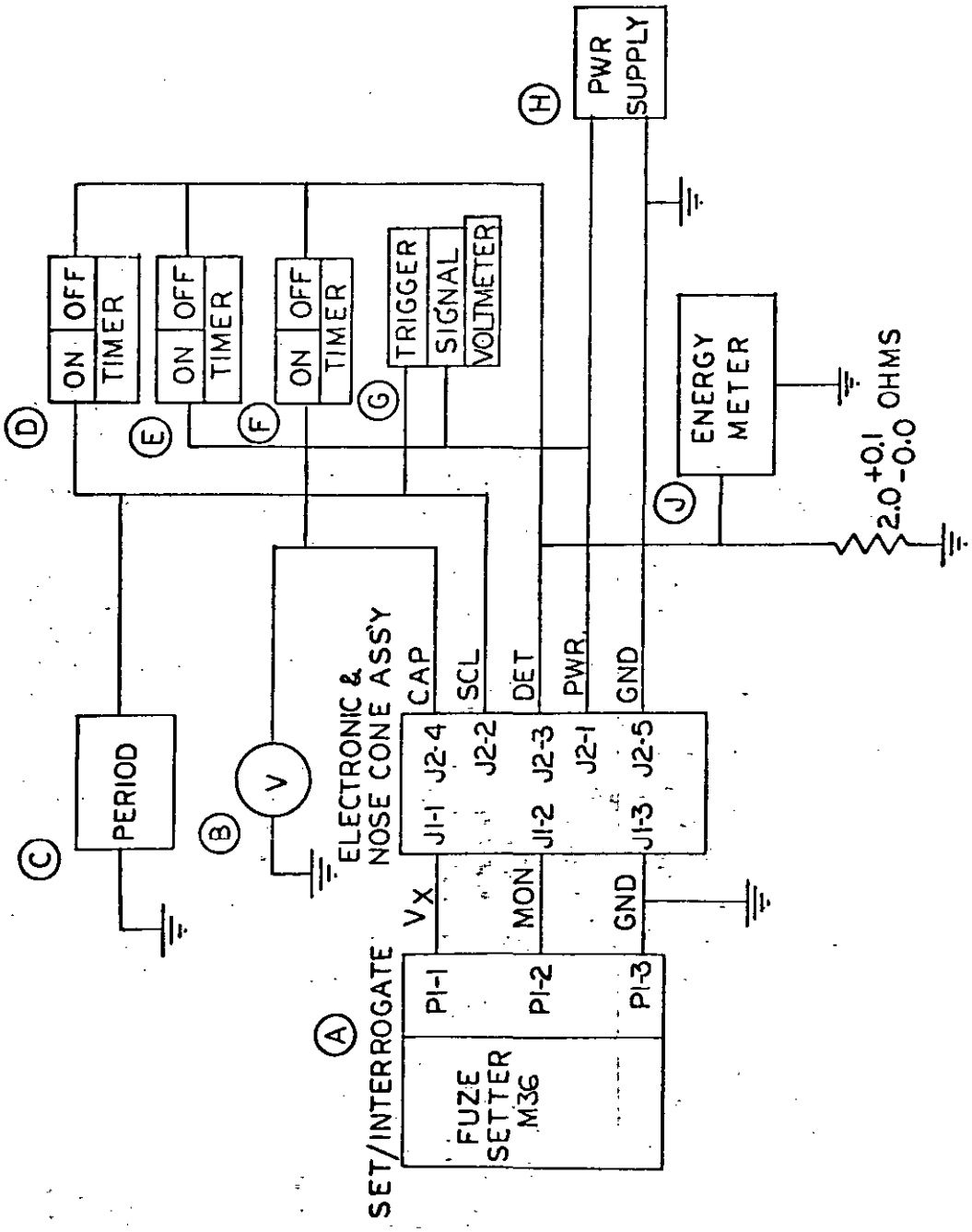


FIGURE-3. SETUP FOR TESTING ELECTRONIC AND NOSE CONE ASS'Y

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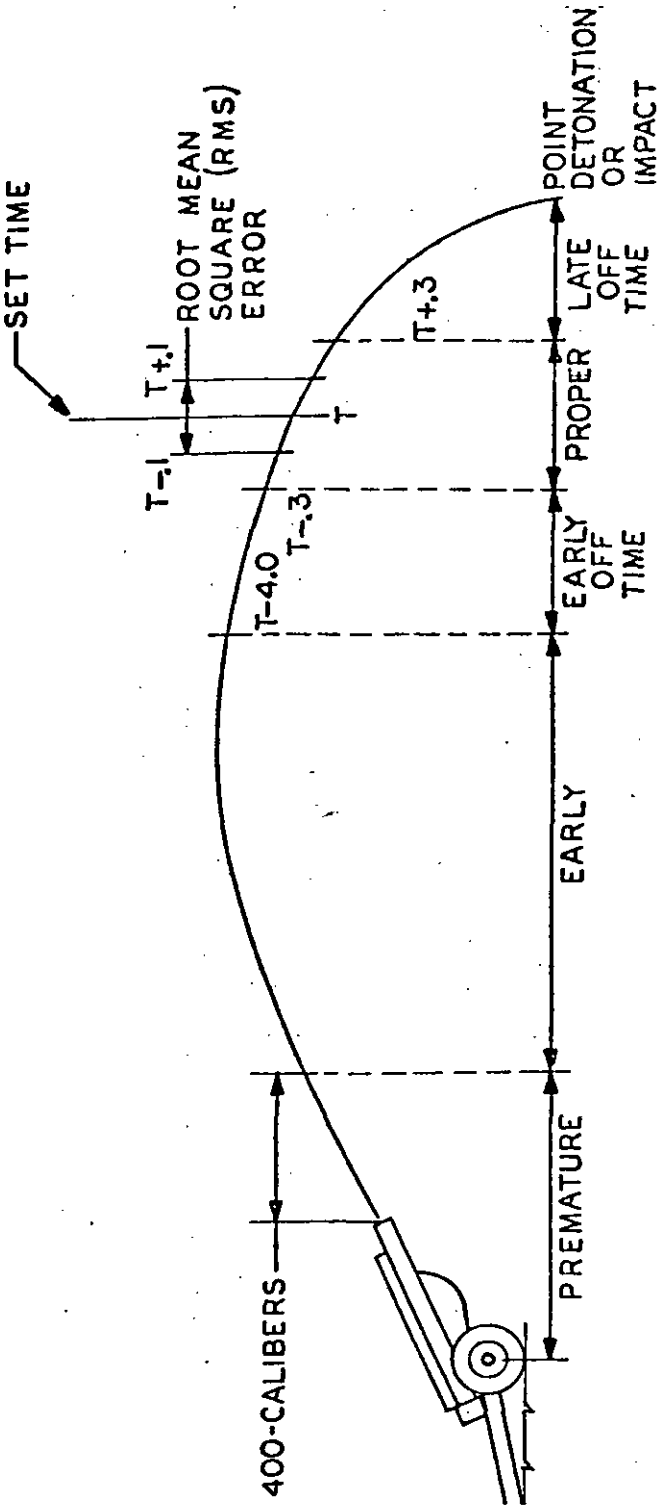


FIGURE 4 TYPES OF BALLISTIC FUNCTIONS

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